Lecture 14 - Neural Networks (I)

ECE 447: Data Analysis and Machine Learning for Engineers

Bahareh Tolooshams, Ph.D.
Assistant Professor,
University of Alberta
btolooshams@ualberta.ca
https://btolooshams.github.io

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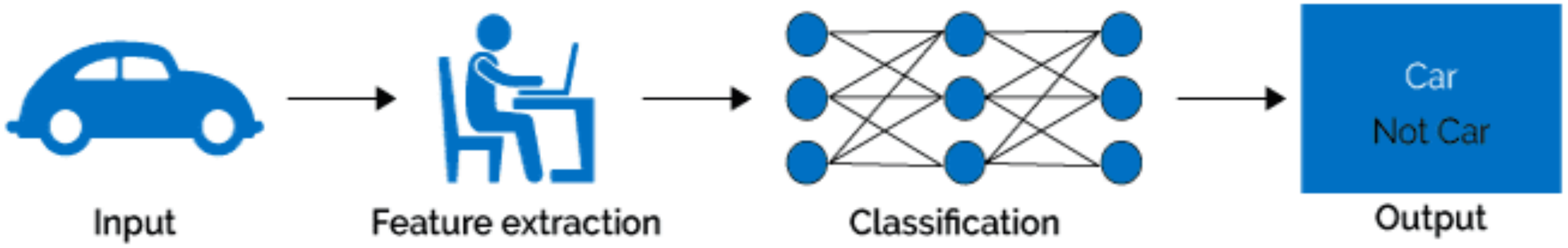
This Lecture Covers ...

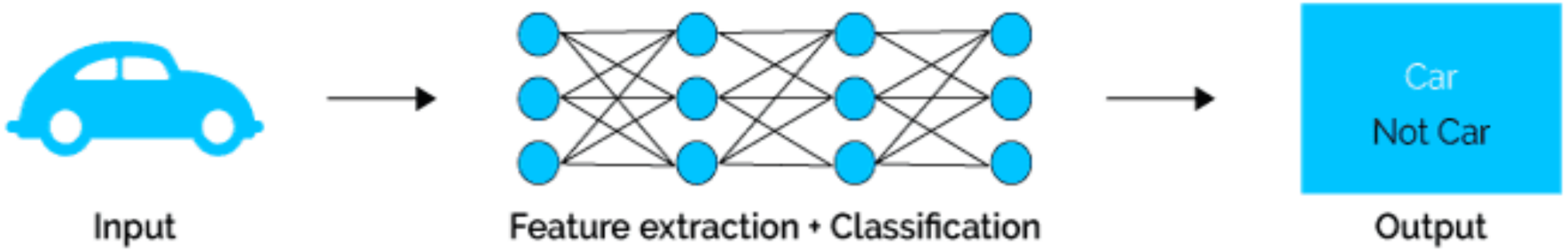
- Machine Learning vs Deep Learning
- Deep Neural Networks Architectures
- Basics of Convlutional Neural Networks

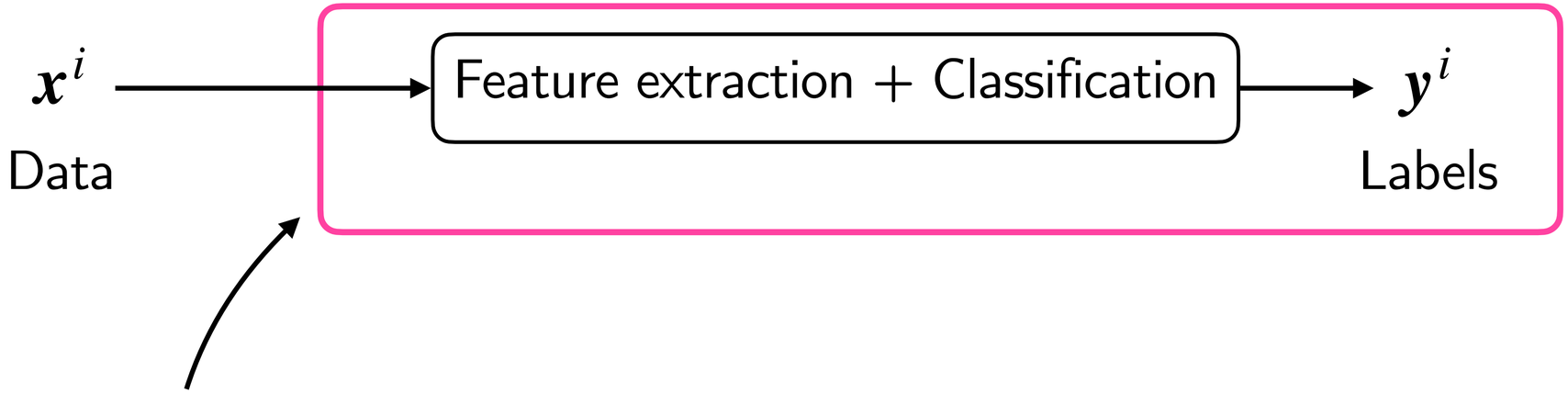
Machine Learning



- *Handcrafted features*
  - *Difficult to design*
  - *Not always optimal*

Deep Learning

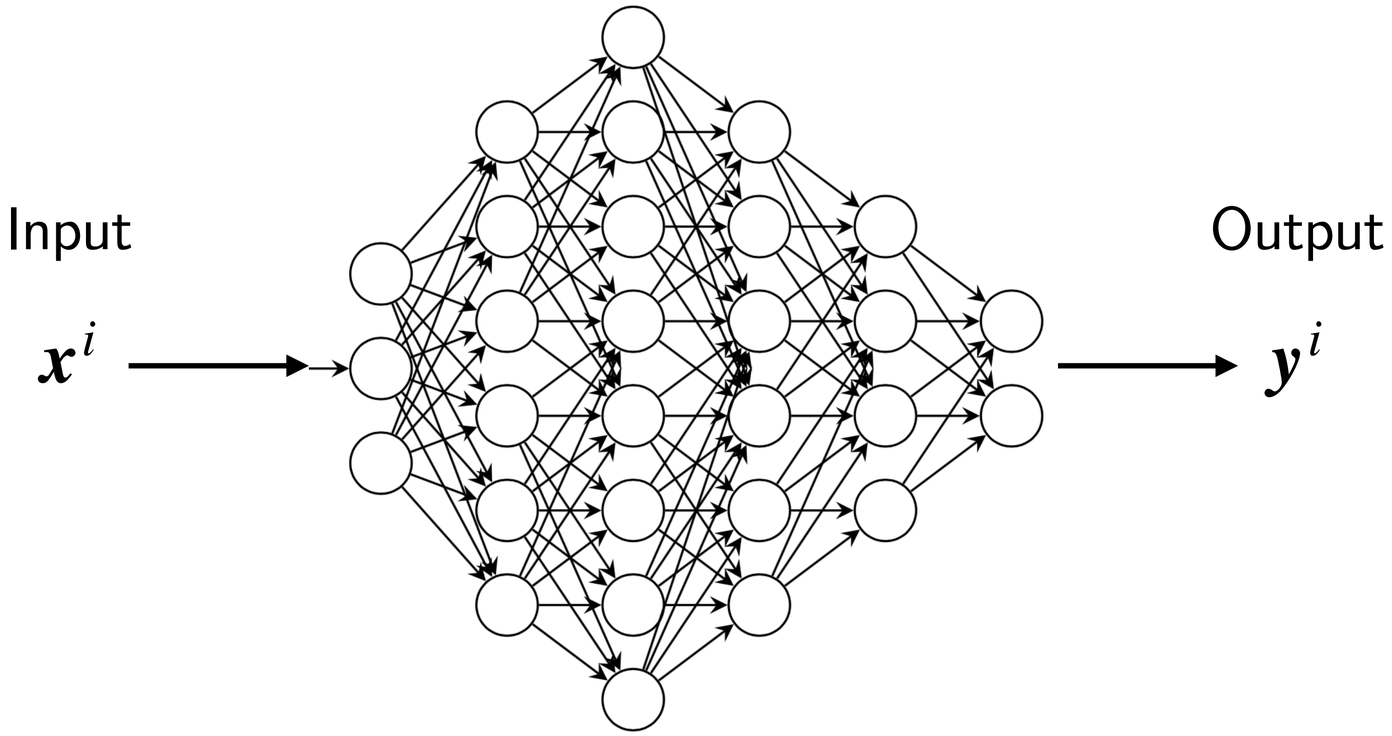




- *Features learned from data*
  - *Automated feature extraction.*
  - *More complicated.*
  - *Requires tuning.*

Deep Learning or Deep Neural Networks

A mapping architecture between your input data and desired output.



What is this mapping $f_\theta(\cdot)$ (neural network)?

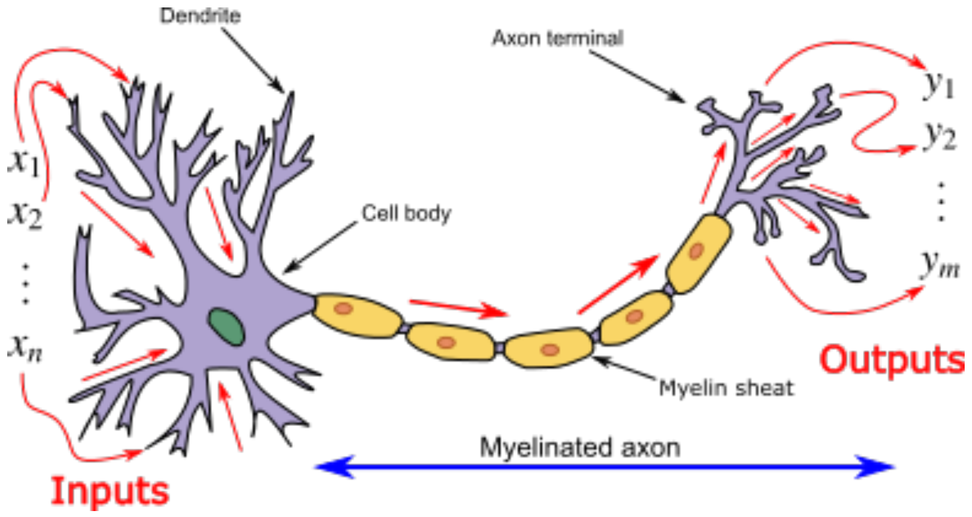
Training loss

$$\min_\theta \sum_{i=1}^{n} \text{Loss}(\boldsymbol{y}^i, \hat{\boldsymbol{y}}^i)$$

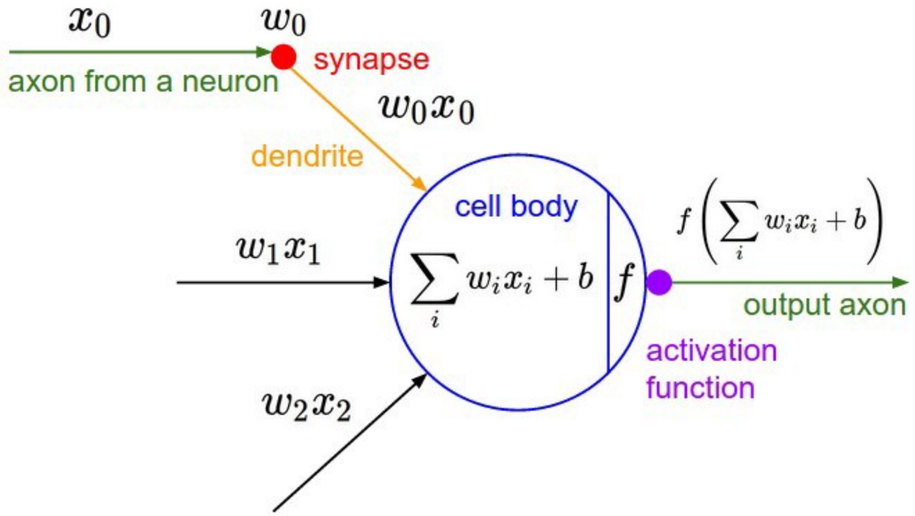
where $\hat{\boldsymbol{y}}^i = f_\theta(\boldsymbol{x}^i)$

Building Block of a Neural Network

Biological neuron



Computational neuron



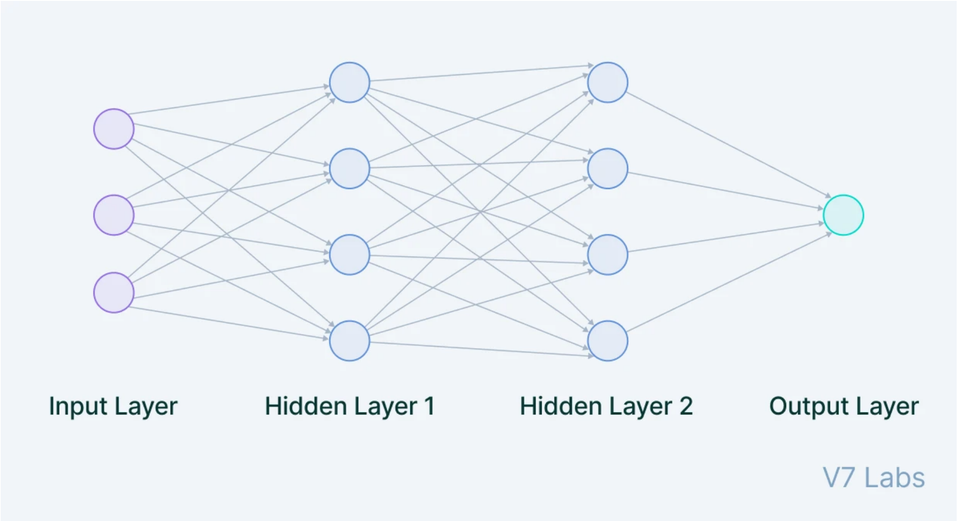
$$\text{output} = \text{nonlinearity}(\boldsymbol{W}\boldsymbol{x} + \boldsymbol{b})$$

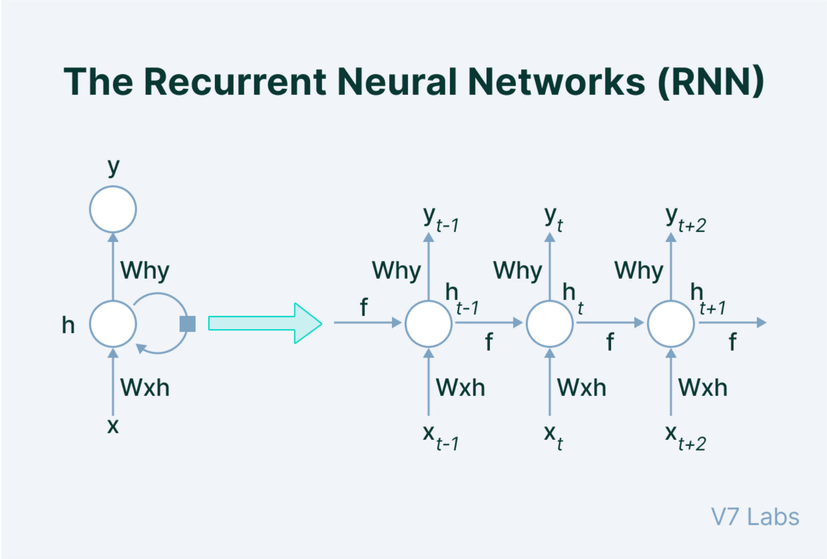
https://www.v7labs.com/blog/neural-network-architectures-guide

Deep Neural Networks (DNNs)

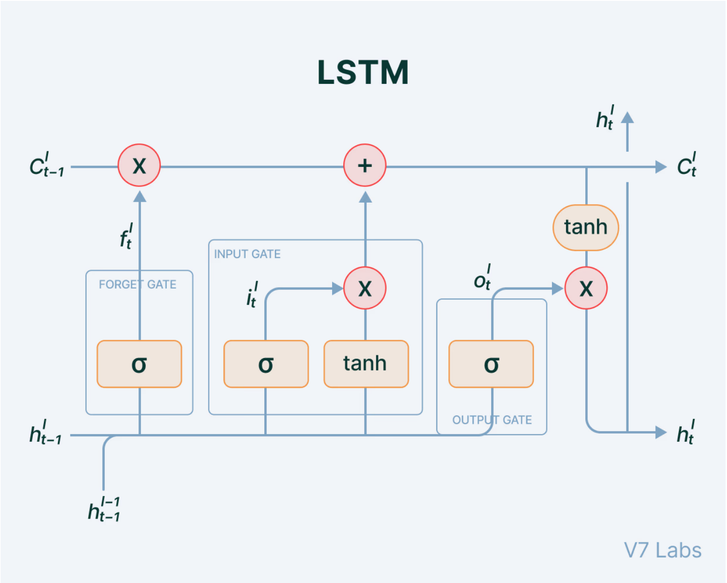
Different networks have different architectures.

Multi-layer Neural Networks or perceptrons (MLP)

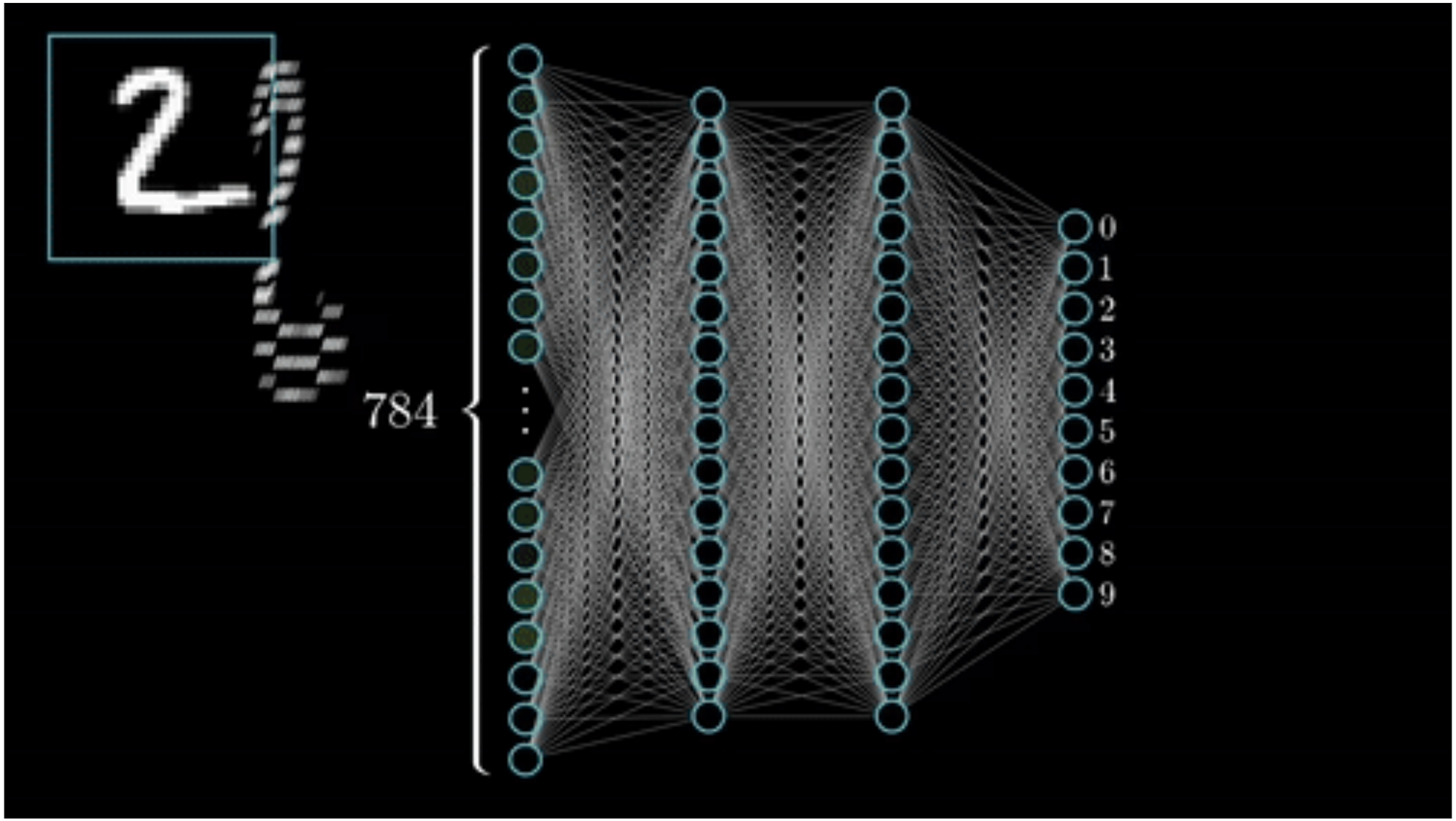




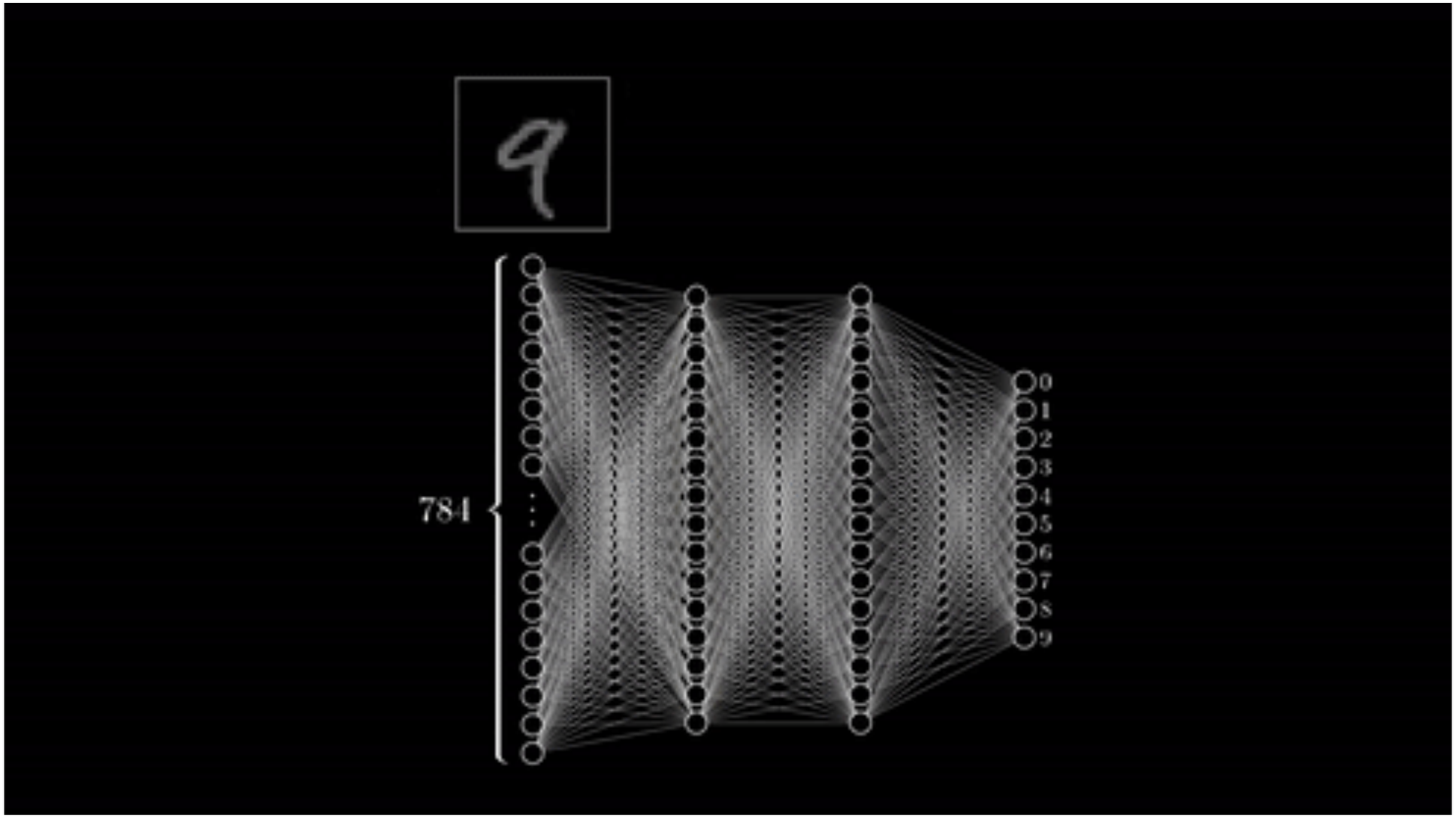
Long Short Term Memory Network



Classification Example with DNNs (I)

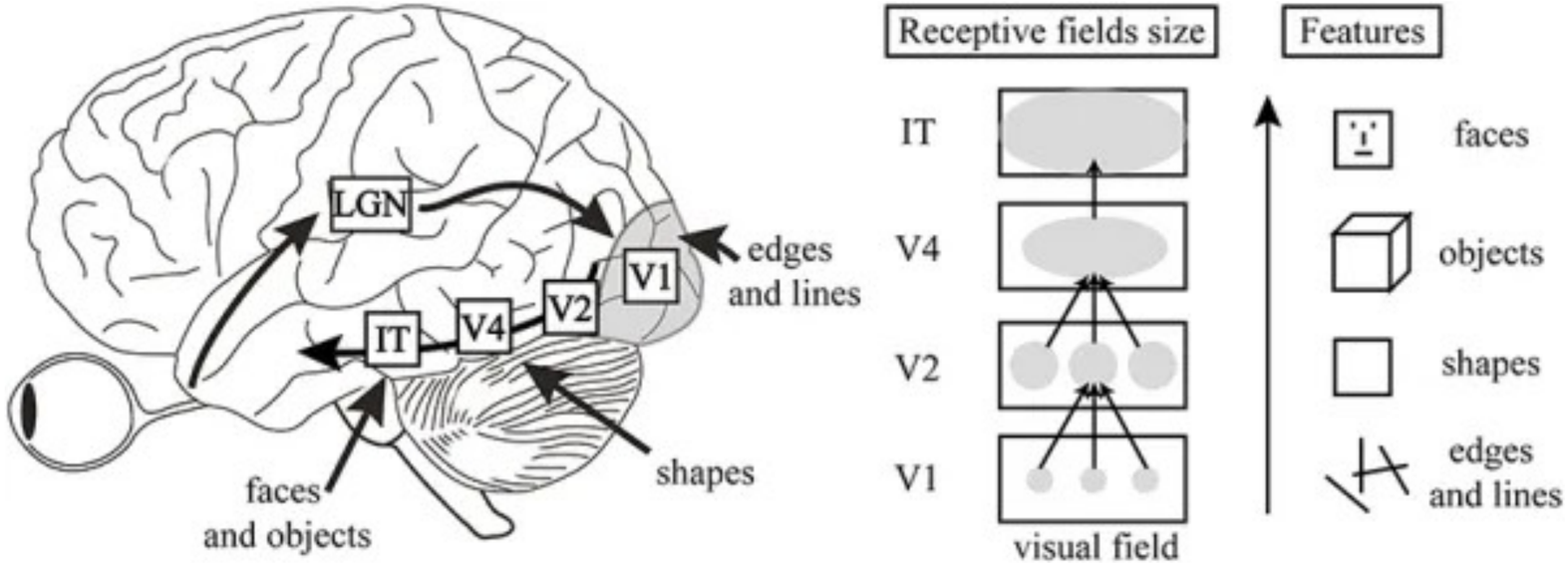


Classification Example with DNNs (II)



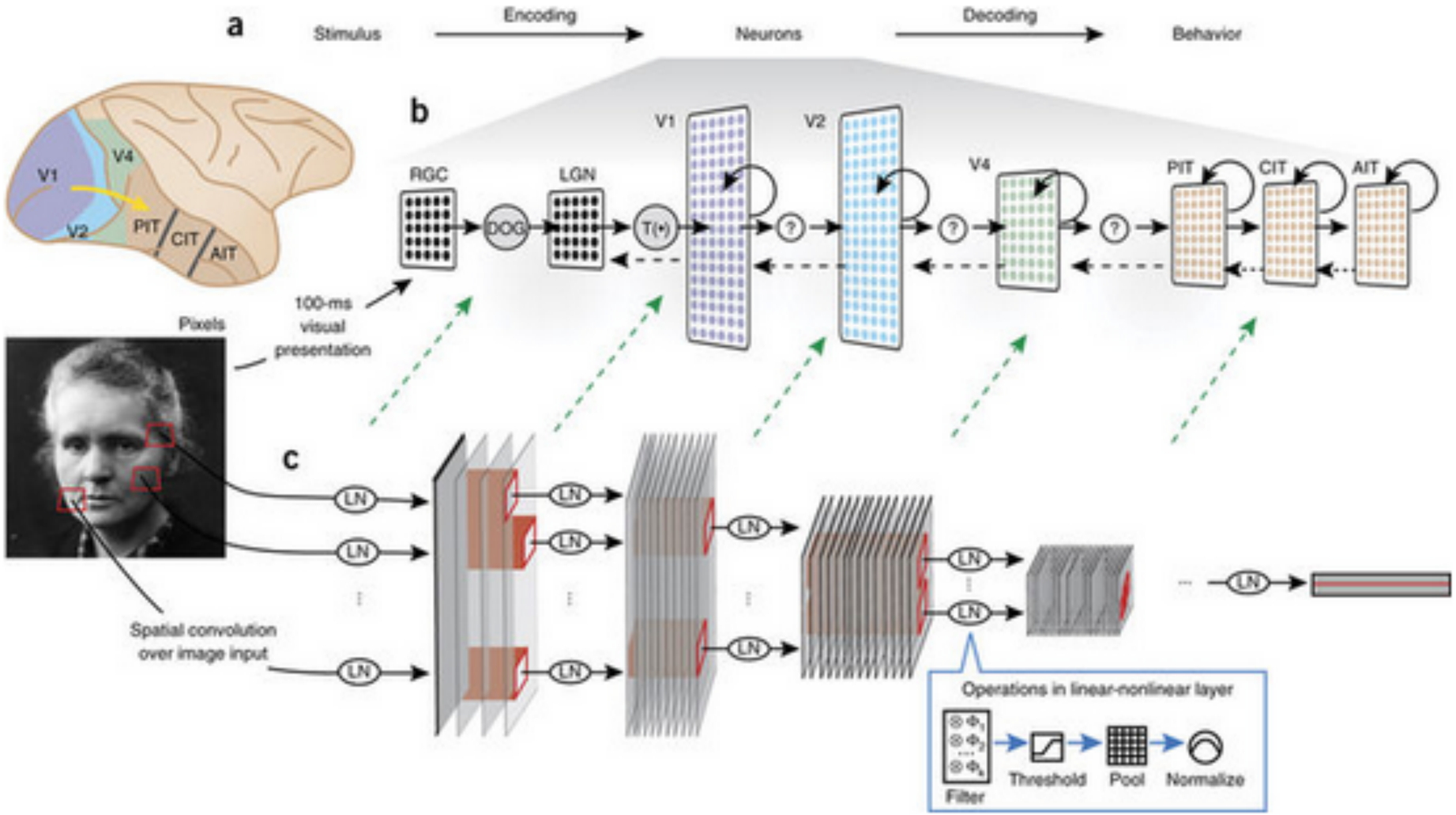
Convolutional Neural Networks (CNNs) (I)

CNNs are inspired by the visual processing system.



https://gracewlindsay.com/2018/05/17/deep-convolutional-neural-networks-as-models-of-the-visual-system-qa/

Convolutional Neural Networks (CNNs) (II)



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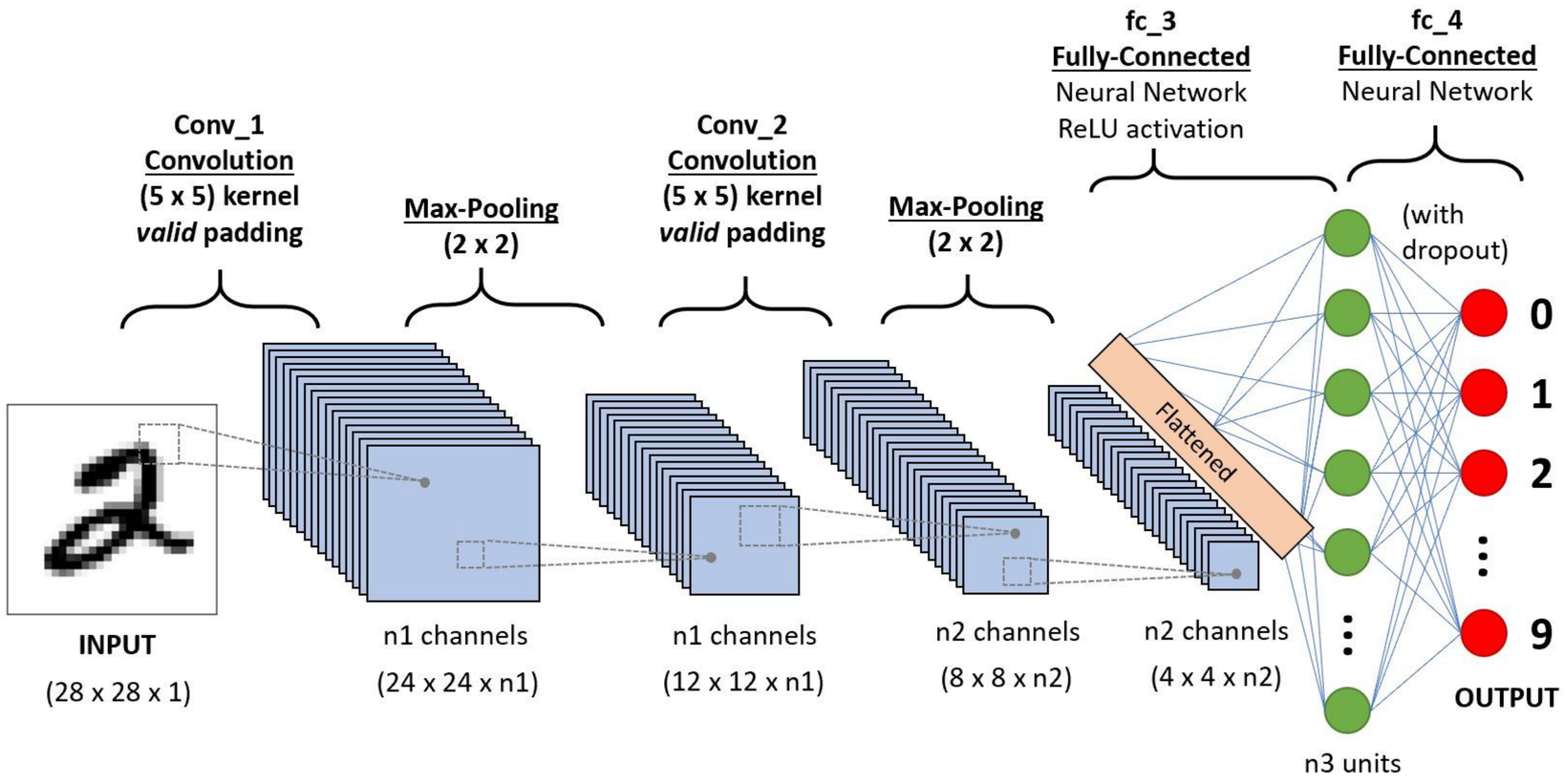
Building Blocks of CNNs

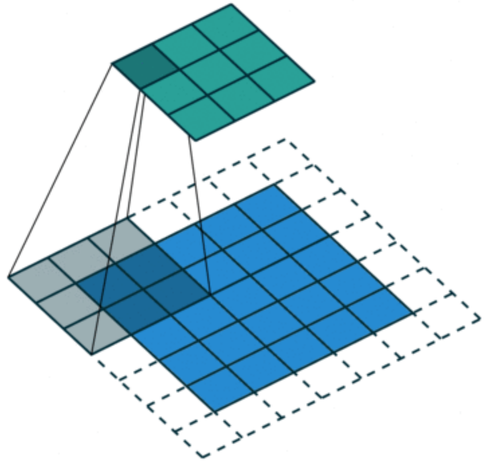
- Convolution
- Max-pooling
- Batch normalization
- Flattening
- Softmax

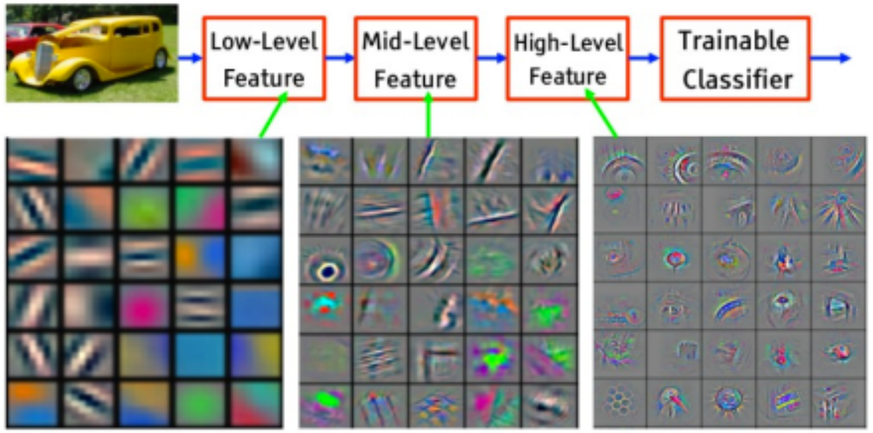


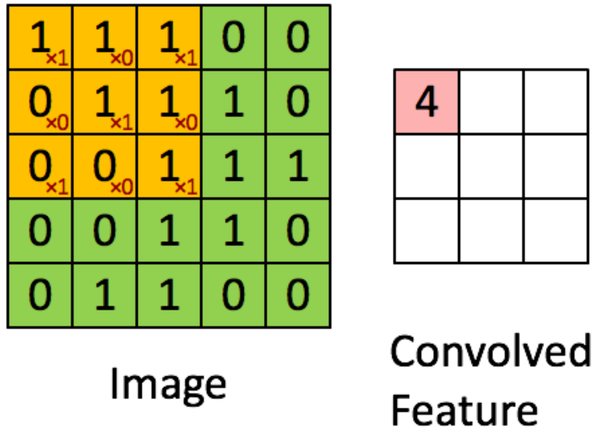
https://medium.com/intro-to-artificial-intelligence/simple-image-classification-using-deep-learning-deep-learning-series-2-5e5b89e97926

CNNs - Convolution

- Performs filtering.
- Neurons respond to local regions (receptive field).
- Inspired by visual sensory information.

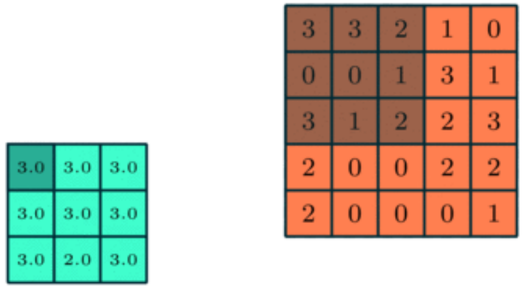


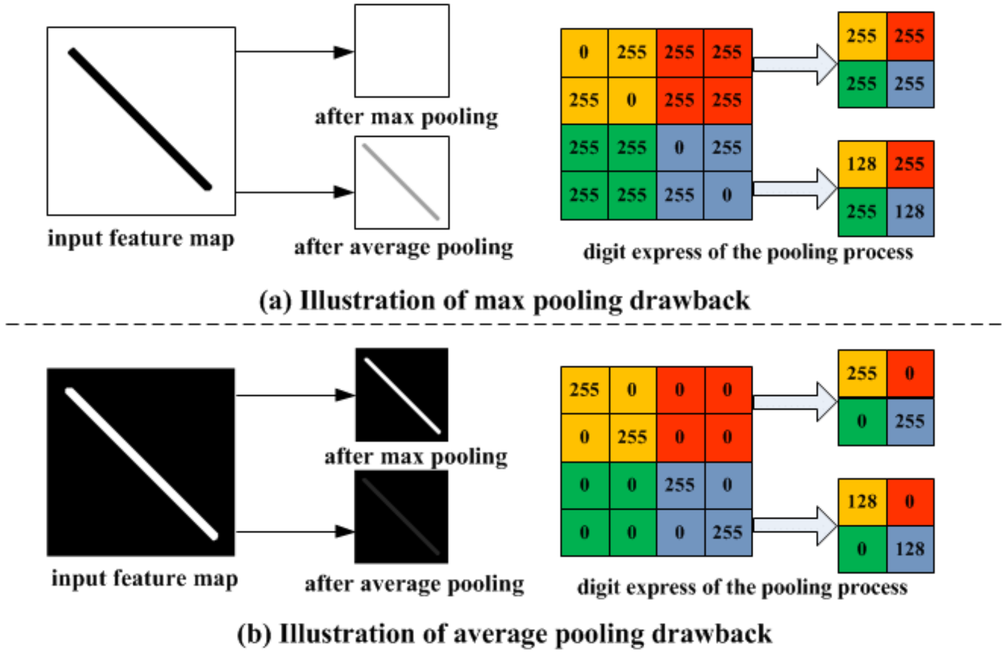


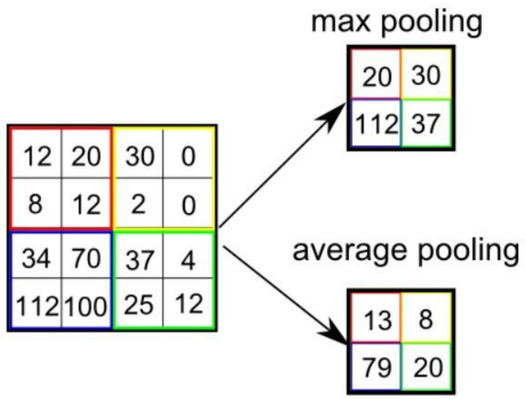


CNNs - Pooling

- Perform a form of down sampling.
- Examples: max pooling, average pooling.

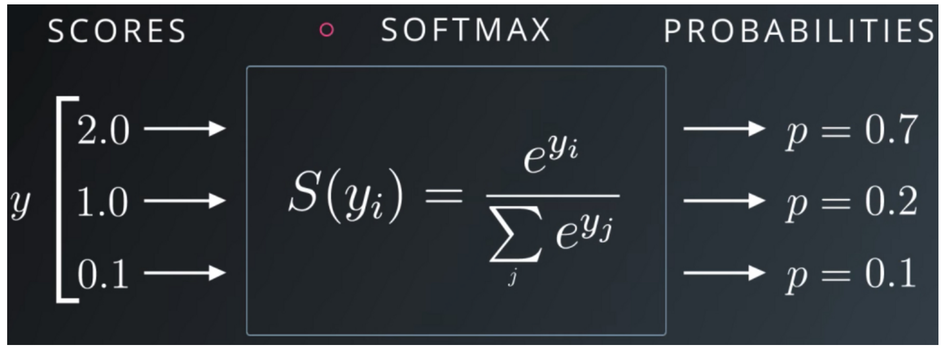


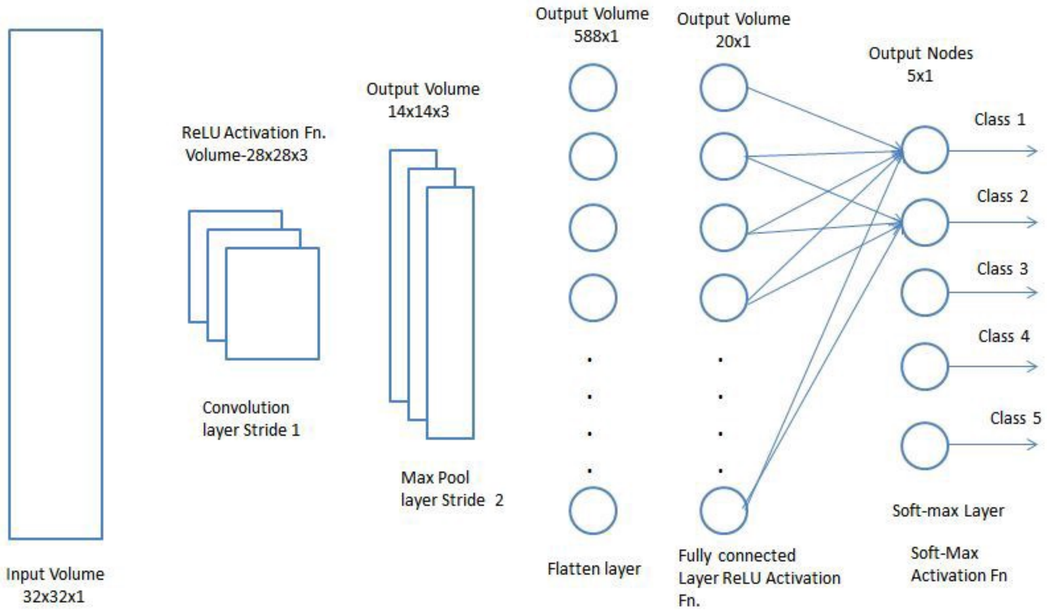




CNNs - Classification layers

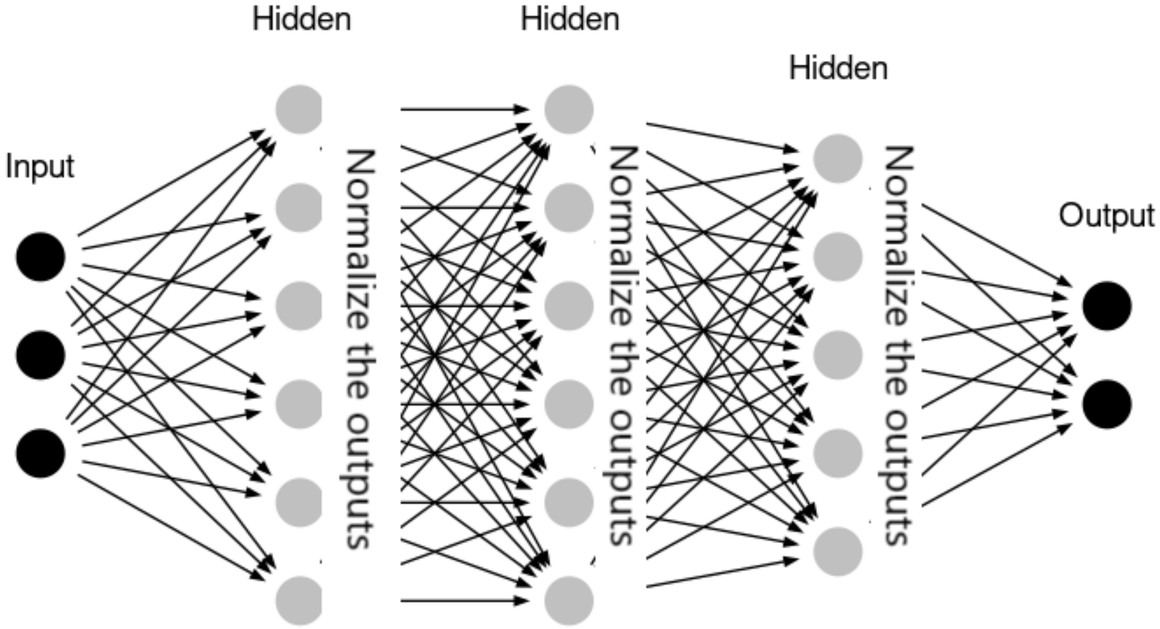
- Flattening: vectorize all the feature maps.
- Fully connected layers.
- Softmax: map into probabilities [0,1].





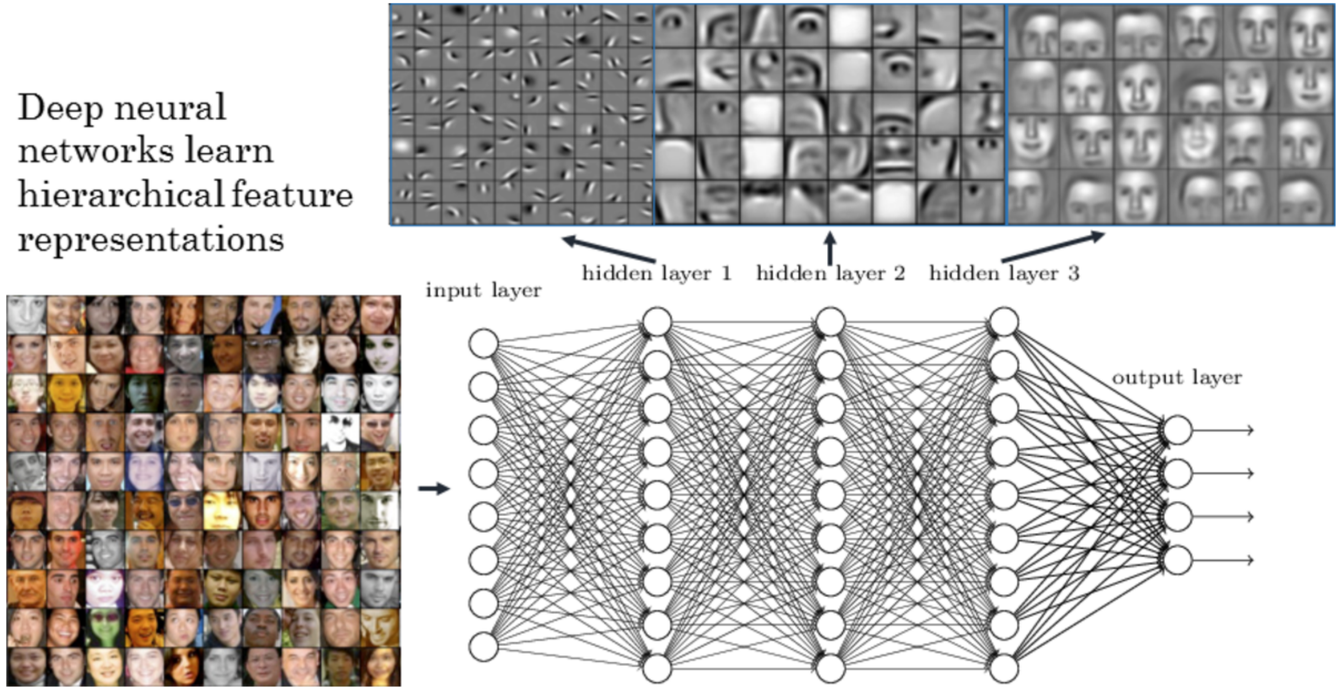
CNNs - Batch Normalization

- Applied to input of each layer.
- Re-center and re-scale after every layer.
- Helps stability and training.



CNNs - Hierarchical Representation

- Low-level features: Edge detectors.
- Deep-level features: Complex and more interpretable by human.



https://www.rsipvision.com/exploring-deep-learning/

Takeaways

- *What are your takeaways?*